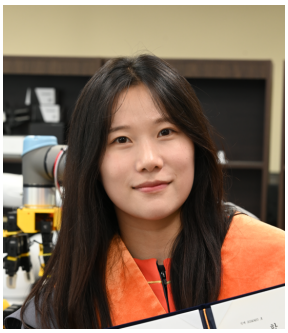
Young-Chae Son

I'm Youngchae Son, a Ph.D. student at Dongguk University in Seoul, working with Prof. Soochul Lim in the Interactive Robotics Lab. My research focuses on robot learning and planning with vision-language foundation models, including Vision-Language Models (VLMs) and Vision-Language-Action (VLA) models. I am particularly interested in enabling robots to make adaptive decisions in real-world environments where task-relevant information may be incomplete, ambiguous, or change during execution. My work explores multimodal perception, task planning, and adaptive manipulation, with an emphasis on integrating and prioritizing sensory observations for long-horizon execution on real robotic systems.



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Education

2025-Present — **Dongguk University** — *Republic of Korea*
Researcher, Mechatronics Engineering
Interactive Robotics Lab

2023-2025 — **Dongguk University** — *Republic of Korea*
M.S. *in Mechatronics Engineering*
Interactive Robotics Lab
Advisor: Prof. Soo-Chul Lim

2018-2023 — **Tech University of Korea** — *Republic of Korea*
B.S. *in Mechatronics Engineering*
2018-2021 Self-Education, Seoul National University of Education, Department of Home Economics

Publications

Journal Article 2026
See Selectively, Act Adaptively: Dual-Level Structural Decomposition for Bimanual Robot Manipulation
Yoon-Ji Choi, **Young-Chae Son**, Soo-Chul Lim*
Under Review

Journal Article 2026
Task-Aware Modality Selection for Multimodal VLA in Robotic Manipulation
Young-Chae Son, Jung-Woo Lee, Yoon-Ji Choi, Dae-Kwan Ko, Soo-Chul Lim*
Under Review

Journal Article 2026
ThermoAct:Thermal-Aware Vision-Language-Action Models for Robotic Perception and Decision-Making
Young-Chae Son, Dae-Kwan Ko, Yoon-Ji Choi, Soo-Chul Lim*
IEEE Robotics and Automation Letters

Conference Paper 2026
A nonmagnetic stepper motor for the automatic evaluation of magnetic field uniformity in Helmholtz coils
Young-Chae Son, Sung-Jung Joo, Po-Gyu Park, Changjin Yun, Soo-Chul Lim*, Hyung-Kew Lee*
IEEE Conference on Precision Electromagnetic Measurements (CPEM)

Journal Article 2026
MulPlanLM: multimodal robotic task planning with vision-language models and physical feedback
Young-Chae Son, DongHan Lee, Soo-Chul Lim*
Intelligent Service Robotics

Conference Paper 2022
The Remote Robotic Arm Control Using Inertial Sensor and Image Recognition
Ki-Jun Han, **Young-Chae Son**, Yang-Hee Lee*
Korean Society for Precision Engineering(KSPE)

Conference Poster 2022
High SNR and High Frame Rate Analog Front-End ICs with Active Stylus for Capacitive Touch Screen Panels
Young-Chae Son, Jae-Sung Ahn*
Europe-Korea Conference on Science and Technology (EKC), Marseille, France

Selected Research Projects

Jul. 2025 – Present
Development of Intelligent Autonomous Manipulation Technology for Humanoids with Reduced Dependency on Real-World Data
Humanoid Manipulation, Robot Learning, Physical AI
Agency: Ministry of Science and ICT (MSIT)
Research on intelligent autonomous manipulation technology for humanoid robots with reduced dependency on real-world data.

Mar. 2025 – Present
Robot Motion Generation AI based on Multimodal Vision/Tactile Information Driven by Language Model
Multimodal Robot Learning, Vision/Tactile Perception, Language Models
Agency: National Research Foundation of Korea (NRF)
Developing robot motion generation methods based on multimodal vision and tactile information driven by language-model-based reasoning.

Aug. 2024 – Dec. 2024
Collecting Large-scale Robot Manipulation Data in Physical Environment
Robot Data Collection, Manipulation, Physical Environment
Agency: National Information Society Agency (NIA)
Collected and organized large-scale robot manipulation data in real-world physical environments for robot learning and automation research.

Honors & Awards

Sep. 2025 – Present
Global Outstanding Talent Scholarship
Full tuition waiver for Ph.D., Dongguk University.

2026
Best Paper Award
BK21FOUR AIMS Center Winter Conference, Dongguk University.

2024
Grand Prize, 1st Place
Transfer Student Counseling Mentoring Testimonial, Tech University of Korea.

2024
Grand Prize, 1st Place
English Presentation Contest, Tech University of Korea.

Patent

Apr. 2026 **Multimodal Robot System for Task-Context-Based Selective Sensory Utilization**
Korean Patent Application, No. 10-2026-0073734.

Apr. 2026 **Method, Apparatus, System, and Computer-Readable Recording Medium for Robot Control Using Thermal Image Information**
Korean Patent Application, No. 10-2026-0067117.

Teaching Experience

Spring 2026 **Teaching Assistant, Autonomous Robotics**
Dongguk University
Autonomous driving practice class using ROS2 and TurtleBot Burger 3

Spring 2025 **Teaching Assistant, Autonomous Robotics**
Dongguk University
Autonomous driving practice class using ROS2 and TurtleBot Burger 3

Fall 2024 **Teaching Assistant, Computational Mechanics**
Dongguk University

Spring 2024 **Teaching Assistant, Autonomous Robotics**
Dongguk University
Autonomous driving practice class using ROS2 and TurtleBot Burger 3

Fall 2023 **Teaching Assistant, Fluid Mechanics**
Dongguk University

Presentations

Oct. 2026 **ThermoAct: Thermal-Aware Vision-Language-Action Models for Robotic Perception and Decision-Making**
IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), Pittsburgh, PA, USA

Jun. 2026 **Selective Perception for Robot: Task-Aware Attention in Multimodal VLA**
Robot Learning Research Group Workshop, Center for Human-Centered Manufacturing Technology, Korea Institute of Industrial Technology (KITECH)

Feb. 2026 **ThermoAct: Thermal-Aware Vision-Language-Action Models for Robotic Perception and Decision-Making**
Locomotion and Manipulation Technology Research Group, Korea Robotics Conference (KRoC), Pyeongchang

Feb. 2026 **ThermoAct: Thermal-Aware Vision-Language-Action Models for Robotic Perception and Decision-Making**
Robot Learning Research Group, Korea Robotics Conference (KRoC), Pyeongchang

Work experience

2022-2023 **Undergraduate Research Intern** *Dongguk University of Korea*
Interactive Robotics Lab
Oct. 2022 - June 2023